

Evaluating Semantic-Layer Effects on Analytical Quality and Efficiency

A controlled paired benchmark across language models

AnswerLayer

Technical report | 14 August 2026

Abstract

This report evaluates whether a governed semantic layer changes analytical answer quality and operational efficiency in a paired benchmark of 9 language models. Each model was evaluated in three independent trials on the same ordered set of 16 expert-gated tasks on a real dataset under two conditions: direct raw-schema access and semantic-layer access. Across the reported models, semantic access increased pooled strict pass rate from 28.7% to 61.3% and reduced SQL calls for 9 of 9 models. After separately pricing uncached input, cache reads, cache writes, and output tokens, estimated model cost declined for 7 of 9 configurations. Retrieval diagnostics further show that component availability and utilization are not sufficient for correctness: observed failures also arise in analytical interpretation, SQL construction, data-limitation handling, relationship resolution, and answer composition. The results support treating the model and semantic layer as a jointly validated production configuration. Three trials per model-condition cell quantify repeatability while remaining descriptive for this fixed task cohort.

Keywords: semantic layer; language-model evaluation; text-to-SQL; retrieval; model efficiency; inference cost

Table 1. Benchmark summary and reporting scope.

Field	Specification
Models	9
Tasks per condition	16
Conditions	Raw schema; semantic layer
Runs in this edition	54 total; three per model-condition pair
Primary outcome	Strict automatic pass rate
Secondary outcomes	Average score, SQL/tool use, elapsed time, token-class cost, retrieval diagnostics
Edition	Public, anonymized

Key findings

Across the matched three-trial benchmark, semantic-layer access improved analytical quality and consistently shortened the path to the database. The magnitude, cost effect, and failure mechanism remained model-dependent.

- **Quality improved for every evaluated model.** Pooled strict pass rate increased from 28.7% under raw-schema access to 61.3% with semantic-layer access. Model-level gains ranged from +6.3 to +58.3 percentage points.
- Database exploration declined for all 9 models. Every configuration issued fewer SQL calls with semantic access, indicating a consistently shorter route from task interpretation to an executable query.

- Estimated model cost decreased for 7 of 9 configurations. The result was not universal because models differed in total token use and in their mix of uncached input, cache reads, cache writes, and output tokens.
- Relevant semantic context being retrieved, resolved, or observed in use was not sufficient for correctness across all models. Residual failures also arose in definition selection, analytical interpretation, SQL construction, data-limitation handling, relationship resolution, and final-answer completeness.
- The same semantic package produced materially different operating patterns. Semantic-mode pass rates ranged from 35.4% to 95.8%, while raw-schema fallback ranged from 14.6% to 91.7% across the reported models.

These results indicate that the model and semantic layer should be qualified as a pair. Monitor strict quality, SQL and tool activity, token-class cost, retrieval behavior, fallback, and failure stage together; no single metric adequately describes the production configuration.

1. Introduction

Semantic layers are intended to reduce the burden of discovering schema structure and reconstructing business logic at query time. Their effect on end-to-end analytical performance, however, depends on how a model retrieves, resolves, interprets, and applies the governed context. This study tests the semantic layer as a system intervention rather than as an isolated documentation asset.

The benchmark addresses three research questions: (RQ1) Does semantic-layer access improve strict analytical answer quality relative to raw-schema access? (RQ2) Does it reduce operational work, including SQL exploration, latency, tokens, and normalized model cost? (RQ3) Where do failures occur when relevant semantic content has been retrieved but the final answer remains incorrect?

The principal contribution is a paired, multi-model comparison that combines outcome metrics with process diagnostics. In addition to pass rate and continuous evaluator score, the report measures SQL activity, token classes, estimated inference cost, semantic retrieval, object resolution, observed use, and raw-schema fallback.

2. Benchmark design

2.1 Experimental conditions

The evaluation used a private analytical dataset and 16 expert-gated tasks. For each model, the ordered task cohort was executed in three independent trials with direct raw-schema access and three with semantic-layer access. The model, dataset, task order, evaluator, prompt family, session mode, timeout, and semantic-layer snapshot were held fixed within each paired comparison.

The semantic layer encodes reusable grains, joins, definitions, formulas, weighting rules, category mappings, and documented data limitations. Expected answers and answer-bearing values remain evaluation evidence and are not semantic-authoring inputs.

2.2 Outcomes and diagnostic metrics

Table 2. Primary outcomes and semantic diagnostic metrics.

Metric	Definition
Strict pass rate	Share of tasks satisfying all required automatic gates.
Average evaluator score	Continuous complementary score; does not override strict pass/fail.

Metric	Definition
Operational activity	SQL calls, failed SQL, tool calls, elapsed time, and recorded token classes.
Normalized model cost	Recorded uncached input, cache-read input, cache-write input, and output tokens priced separately using the edition-specific rate card.
Resolved-object utilization	Observed used objects divided by resolved objects.
Discovery precision	Observed used objects divided by retrieved objects.
Payload utilization	Estimated input tokens for used semantic payload divided by estimated input tokens for resolved semantic payload.
Fallback	Share of semantic-mode attempts whose trace records raw-schema inspection.

2.3 Evaluation rubric and pass criteria

Every case required successful inquiry execution. The answer was then assessed through one or more case-level criteria selected for the form of the expected result. Typed value checks were used where the reference answer could be stated precisely; an LLM judge was reserved for the two cases whose correct outcome was a qualified data-limitation conclusion. The typed categories overlap: a single case may require both numbers and identifiers, and every required assertion must pass.

- **Exact-value matching.** Thirteen cases contained 42 typed numeric assertions. Eleven assertions required exact equality; 31 used a specified absolute tolerance. Percentage assertions retained their unit semantics so sign, magnitude, and scale remained part of correctness rather than being reduced to unrestricted text matching.
- **Identifier matching.** Eleven cases contained 25 required identifiers. The matcher checked canonical names and, where configured, accepted explicit aliases; ten identifier assertions included an alias. This allowed recognized abbreviations without weakening the requirement to return the requested entities.
- **LLM-as-judge.** Two cases used a fixed judge rubric because correctness depended on recognizing that the requested result was not reliably supported by the available data. The judge assessed whether the answer communicated the central limitation, avoided presenting an undocumented proxy as fact, and described the evidence or clarification needed; exact wording was not required.

Strict pass rate is conjunctive: successful execution and every required case criterion must pass. The reported average evaluator score is therefore complementary to, and does not override, strict pass/fail.

2.4 Analysis and interpretation

Comparisons are paired descriptively within model and repetition. Pass rates pool the 48 task attempts in each model-condition cell. Observed variability is summarized from the three trial-level values using the sample standard deviation (denominator $n - 1$). Percentage-point changes are used for pass rate; paired proportional changes are calculated within repetition for SQL and tokens. Qualitative trace review assigns each failure to its primary process stage without altering the recorded automatic outcome.

Whiskers in Figures 1-3 denote the mean plus or minus one sample standard deviation across the three trials. They are descriptive dispersion summaries, not standard errors, confidence intervals, or significance tests. With $n=3$ and a fixed task cohort, the analysis characterizes observed run-to-run variability but does not support population-level inference or generalization beyond this benchmark.

2.5 Execution errors and missing usage

The reported matrix contains 8 execution errors across 864 task attempts (0.9%). Seven were 300-second timeouts; the remaining error was malformed model output. Errors remain observed non-passes and retain their measured duration. Missing additive usage fields are estimated from successful repetitions of the identical model, condition, and task; quality, reliability, and latency are never imputed.

3. Results

3.1 Answer quality

With all other tested conditions held constant, all 9 models passed more tasks when the semantic layer was enabled. Pooled strict pass rate increased from 28.7% to 61.3%, a 32.6-percentage-point lift. Model-level improvements ranged from +6.3 to +58.3 percentage points, showing a common direction but materially different effect sizes.

Across models, the observed sample standard deviation of trial-level pass rate ranged from 3.6 to 9.5 percentage points under raw-schema access and from 0.0 to 9.5 under semantic-layer access. These values describe dispersion among the three benchmark trials; they do not quantify uncertainty in a population parameter.

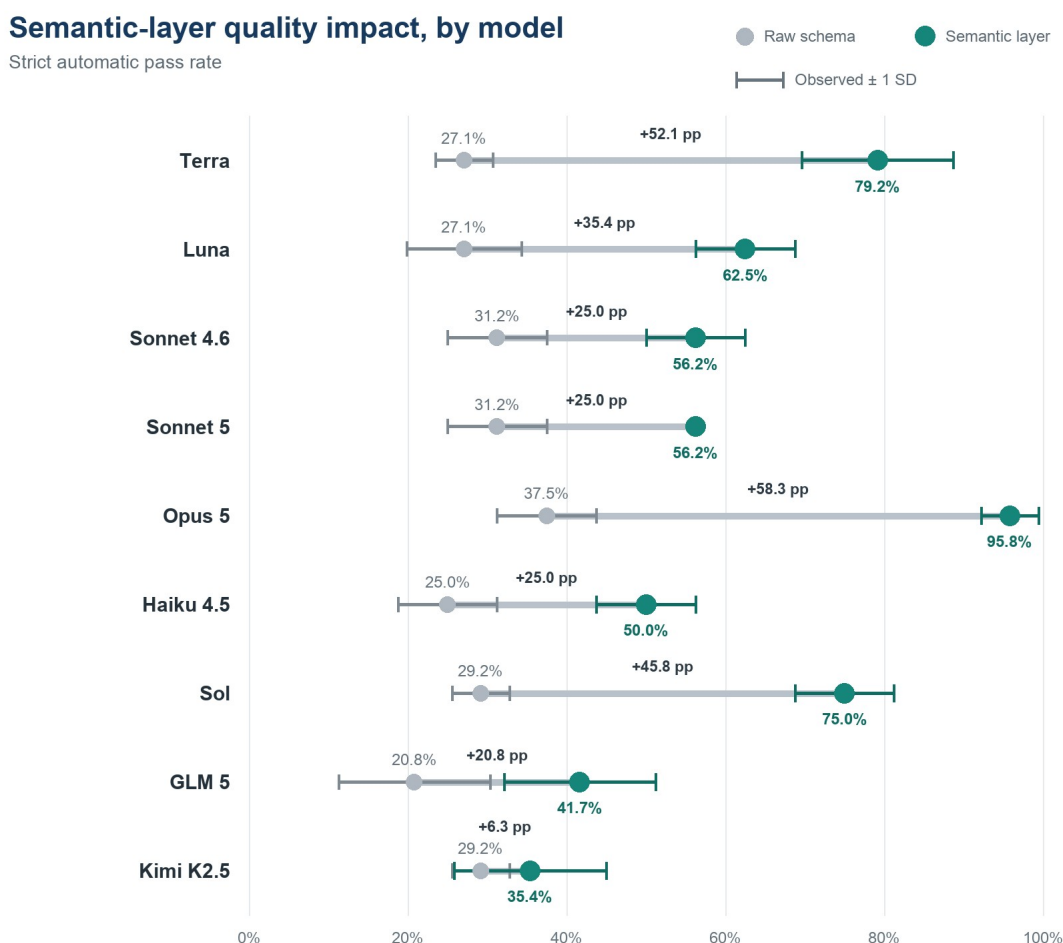


Figure 1. Mean strict automatic pass rate across three matched trials per condition. Points equal the pooled rate across 48 task attempts; whiskers show plus or minus one sample standard deviation of the three trial-level rates.

Table 3. Paired quality and efficiency outcomes by model.

Model	Pass rate mean $\pm$ SD	Δ quality	Avg. score mean $\pm$ SD	SQL reduction mean $\pm$ SD	Token change mean $\pm$ SD
Terra	R 27.1 $\pm$ 3.6% S 79.2 $\pm$ 9.5%	+52.1 pp	R 75.41 $\pm$ 2.09 S 93.72 $\pm$ 2.86	75.9 $\pm$ 0.8%	+5.6 $\pm$ 10.4%
Luna	R 27.1 $\pm$ 7.2% S 62.5 $\pm$ 6.2%	+35.4 pp	R 77.51 $\pm$ 3.14 S 89.02 $\pm$ 2.01	67.9 $\pm$ 6.7%	+14.6 $\pm$ 22.0%
Sonnet 4.6	R 31.2 $\pm$ 6.2% S 56.2 $\pm$ 6.2%	+25.0 pp	R 80.68 $\pm$ 0.86 S 87.29 $\pm$ 1.31	37.6 $\pm$ 9.8%	+7.8 $\pm$ 6.9%
Sonnet 5	R 31.2 $\pm$ 6.2% S 56.2 $\pm$ 0.0%	+25.0 pp	R 73.84 $\pm$ 1.28 S 87.69 $\pm$ 5.26	59.5 $\pm$ 6.8%	-19.3 $\pm$ 13.0%
Opus 5	R 37.5 $\pm$ 6.2% S 95.8 $\pm$ 3.6%	+58.3 pp	R 86.22 $\pm$ 1.44 S 99.41 $\pm$ 0.51	67.6 $\pm$ 3.0%	-44.2 $\pm$ 4.1%
Haiku 4.5	R 25.0 $\pm$ 6.2% S 50.0 $\pm$ 6.2%	+25.0 pp	R 73.15 $\pm$ 3.87 S 85.25 $\pm$ 0.85	42.7 $\pm$ 23.4%	-16.6 $\pm$ 32.7%
Sol	R 29.2 $\pm$ 3.6% S 75.0 $\pm$ 6.2%	+45.8 pp	R 75.56 $\pm$ 1.71 S 91.77 $\pm$ 4.02	83.7 $\pm$ 1.9%	-14.1 $\pm$ 9.1%
GLM 5	R 20.8 $\pm$ 9.5% S 41.7 $\pm$ 9.5%	+20.8 pp	R 75.38 $\pm$ 1.85 S 81.78 $\pm$ 0.70	35.1 $\pm$ 12.0%	+16.9 $\pm$ 24.0%
Kimi K2.5	R 29.2 $\pm$ 3.6% S 35.4 $\pm$ 9.5%	+6.3 pp	R 76.45 $\pm$ 1.91 S 80.39 $\pm$ 0.58	42.8 $\pm$ 4.9%	-8.3 $\pm$ 12.0%

Note: R = raw schema; S = semantic layer. SD is the sample standard deviation across three trial-level values. Pass rate is the strict recorded evaluator outcome; average score is complementary and does not override pass/fail gates. SQL and token columns summarize paired within-repetition percentage changes.

3.2 Operational efficiency

Semantic-layer access reduced direct database exploration for every evaluated model. Aggregate SQL calls fell for all 9 models and aggregate run duration fell for 9 of 9. Estimated model cost per case declined for 7 of 9 configurations after pricing each recorded token class at the edition-specific rate card.

Efficiency moved in different ways

Fewer SQL calls were universal; cost effects were model-dependent

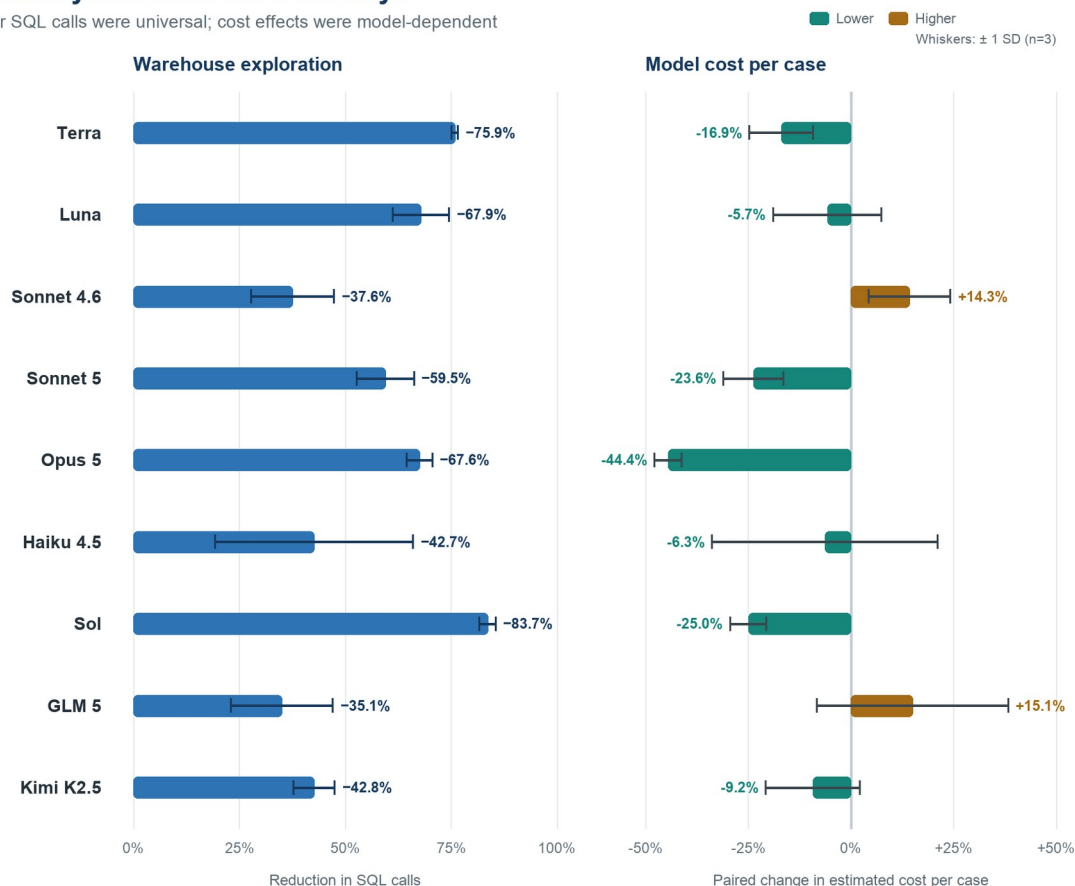


Figure 2. Mean paired within-repetition SQL-call reduction (left) and percentage change in estimated model cost per evaluated case (right). Bars summarize the three matched raw-schema-to-semantic-layer changes for each model; whiskers show plus or minus one sample standard deviation. Negative cost changes indicate lower estimated cost with semantic-layer access. These are observed trial dispersions, not confidence intervals.

3.2.1 Interpretation of efficiency metrics

- **Warehouse work:** mean paired SQL-call reductions ranged from 35.1% to 83.7%. For every model, the mean minus one observed SD remained above zero.
- **Cost per case:** mean paired changes ranged from -44.4% to +15.1%; the observed mean plus or minus one SD crossed zero for 4 of 9 models. This is a descriptive variability check, not a significance test.
- **Tool activity:** 9 of 9 models used fewer tools in aggregate despite universal reductions in SQL queries.
- **Elapsed time:** aggregate duration declined for every model, ranging from 3.3% to 35.2%.

Taken together, these measures indicate that semantic access consistently shortened the path to the database, while its effect on normalized model cost depended on the model's trajectory and mix of uncached input, cache reads, cache writes, and output.

3.3 Normalized model cost

Repricing the recorded token classes at current direct-provider API list rates lowers estimated model spend for 7 of 9 configurations. Cache reads receive the applicable discounted rate; cache writes receive the applicable premium; uncached input and output use their own rates.

Cost versus score

Raw-schema to semantic-layer movement; USD list-price estimate per case

○ Raw schema ● Semantic layer

Whiskers: observed ± 1 SD across 3 trials

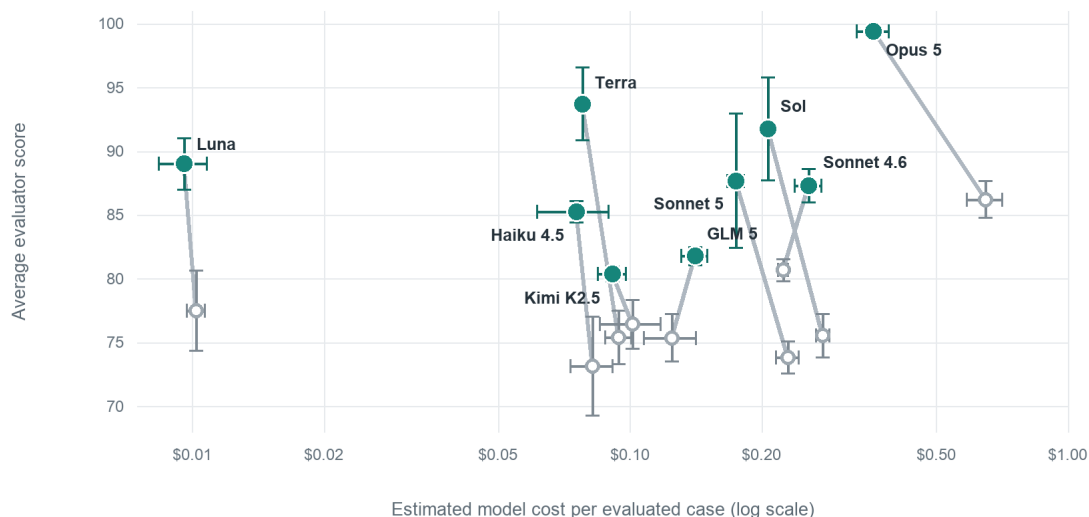


Figure 3. Trial-mean evaluator score versus trial-mean estimated model cost per case. Horizontal and vertical whiskers show plus or minus one sample standard deviation across three trials; lines connect each model's raw-schema and semantic-layer means. The cost axis is logarithmic.

Table 4. Estimated inquiry-model cost per case and evaluator score, including observed trial variability.

Model	Raw cost / case mean $\pm$ SD	Semantic cost / case mean $\pm$ SD	Δ cost	Score mean $\pm$ SD
Terra	\$0.094 $\pm$ \$0.0065	\$0.078 $\pm$ \$0.0019	-17.3%	R 75.41 $\pm$ 2.09 S 93.72 $\pm$ 2.86
Luna	\$0.0102 $\pm$ \$0.0005	\$0.0096 $\pm$ \$0.0012	-5.9%	R 77.51 $\pm$ 3.14 S 89.02 $\pm$ 2.01
Sonnet 4.6	\$0.224 $\pm$ \$0.0057	\$0.255 $\pm$ \$0.0179	+14.2%	R 80.68 $\pm$ 0.86 S 87.29 $\pm$ 1.31
Sonnet 5	\$0.229 $\pm$ \$0.0139	\$0.174 $\pm$ \$0.0079	-23.9%	R 73.84 $\pm$ 1.28 S 87.69 $\pm$ 5.26
Opus 5	\$0.648 $\pm$ \$0.060	\$0.360 $\pm$ \$0.030	-44.5%	R 86.22 $\pm$ 1.44 S 99.41 $\pm$ 0.51
Haiku 4.5	\$0.082 $\pm$ \$0.0092	\$0.075 $\pm$ \$0.0139	-8.3%	R 73.15 $\pm$ 3.87 S 85.25 $\pm$ 0.85
Sol	\$0.276 $\pm$ \$0.0097	\$0.207 $\pm$ \$0.0051	-25.1%	R 75.56 $\pm$ 1.71 S 91.77 $\pm$ 4.02
GLM 5	\$0.124 $\pm$ \$0.0168	\$0.141 $\pm$ \$0.0097	+13.0%	R 75.38 $\pm$ 1.85 S 81.78 $\pm$ 0.70
Kimi K2.5	\$0.101 $\pm$ \$0.0160	\$0.091 $\pm$ \$0.0067	-10.2%	R 76.45 $\pm$ 1.91 S 80.39 $\pm$ 0.58

In Figure 3, movement upward and leftward represents a higher evaluator score at lower estimated model cost. Cost dispersion is recomputed from each trial's recorded token classes using the edition-specific rate card. Whiskers describe observed variability only and should not be interpreted as confidence regions. Cost estimates use direct-provider API list rates retrieved 13 August 2026; token classes are priced separately where reported, and model-specific sources are listed in the references.

3.4 Retrieval and utilization

The semantic layer can be present without being decisive. The diagnostic funnel separates what was retrieved, what was resolved into detailed context, what was observed in use, and whether the model still fell back to raw schema. Rates are pooled from case-level totals across the three semantic trials (48 attempts per model), rather than averaged from case percentages. A semantic object can contribute once per attempt in which it appears. Where an execution error omitted diagnostics, the adjusted total includes the prespecified donor estimate described in Section 2.5.

- **Resolved-object utilization.** The number of case-level unique semantic objects attributed as used in the trace divided by the number resolved into detailed semantic context. It measures use after resolution, not whether the object was the correct choice or was applied correctly.
- **Discovery precision.** The number of case-level unique semantic objects attributed as used divided by the number surfaced during discovery. Because the denominator includes retrieved objects that were never resolved, it measures end-to-end retrieval selectivity rather than answer correctness.
- **Payload utilization.** The estimated input-token weight of used semantic payload divided by the estimated input-token weight of resolved semantic payload, pooled across attempts. This is a diagnostic estimate of useful context share, not a provider-billed token count.
- **Fallback.** The number of semantic-mode attempts whose trace records raw-schema inspection divided by all 48 semantic-mode attempts for the model. Fallback can coexist with semantic-object use and does not by itself indicate a failed answer.

Retrieval is not the same as utilisation

Semantic-mode funnel rates; higher utilisation is useful, but not sufficient for correctness

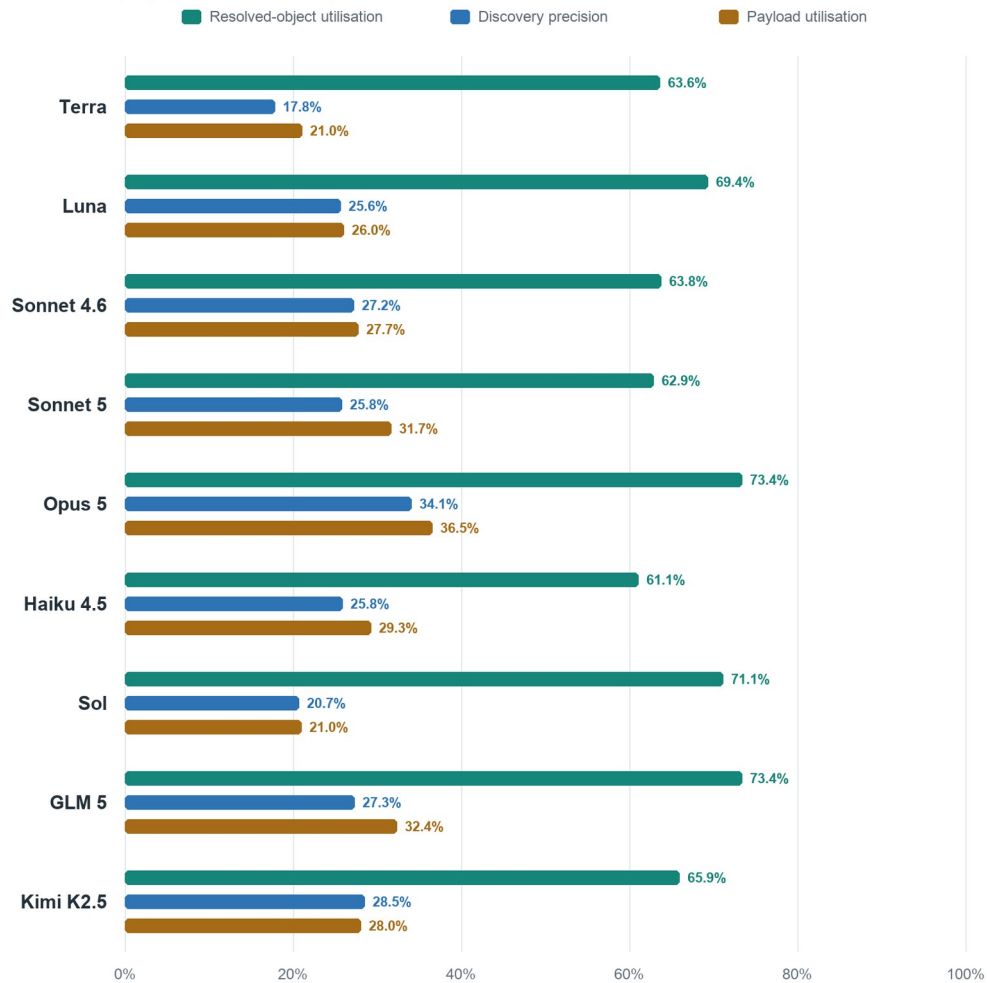


Figure 4. Three different semantic-mode rates: used / resolved objects; used / retrieved objects; and estimated input-token weight for used semantic payload / estimated input-token weight for resolved semantic payload.

Table 5. Semantic-condition retrieval, observed use, and fallback by model.

Model	Retrieved → resolved → used	Resolved use	Discovery	Payload use	Fallback cases
Terra	1023 → 286 → 182	63.6%	17.8%	21.0%	37/48
Luna	1156 → 428 → 296	69.4%	25.6%	26.0%	24/48
Sonnet 4.6	808 → 345 → 220	63.8%	27.2%	27.7%	13/48
Sonnet 5	999 → 410 → 258	62.9%	25.8%	31.7%	42/48
Opus 5	930 → 432 → 317	73.4%	34.1%	36.5%	7/48
Haiku 4.5	844 → 357 → 218	61.1%	25.8%	29.3%	33/48
Sol	1243 → 362 → 257	71.1%	20.7%	21.0%	19/48
GLM 5	882 → 328 → 241	73.4%	27.3%	32.4%	33/48
Kimi K2.5	914 → 395 → 260	65.9%	28.5%	28.0%	44/48

3.4.1 Cross-model interpretation

- **High utilization is not sufficient for correctness.** GLM used 73.4% of resolved objects, yet its semantic pass rate was 41.7%; component choice, SQL construction, and answer composition still matter.
- Strong use with low fallback can shift failures downstream. Opus 5 combined 73.4% resolved-object utilization with only 14.6% fallback and a 95.8% pass rate; its two semantic misses were evidence-completeness failures rather than retrieval failures.
- **Fallback remains model-specific.** Sonnet 5 fell back to raw schema in 87.5% of semantic attempts despite using 62.9% of resolved objects, whereas Opus 5 used a similar share with much less fallback and substantially higher quality.
- The same semantic package produced markedly different operating patterns: semantic pass rates ranged from 35.4% to 95.8%, and fallback ranged from 14.6% to 91.7%.

These patterns imply that no single utilization rate is a proxy for correctness. Retrieval, resolution, observed use, fallback, failed SQL, and final-answer validation must be interpreted jointly.

4. Failure analysis

The traces show a recurring pattern: relevant semantic context is often available, but availability does not guarantee the model will select the canonical definition, translate it into the right query, or preserve the required evidence in the answer. Common failures therefore span the full path from retrieval to evaluation.

Table 6. Recurring failure classes and diagnostic signatures.

Common cause	What goes wrong	Typical trace signature
Analytical interpretation	A relevant component is found, but the model chooses a competing convention, adds an unsupported condition, or selects the wrong measure.	High semantic use with a wrong cohort, grain, window, or aggregation.
Query construction and scope	The intended definition is not carried faithfully into SQL through filters, joins, weighting, grouping, or retry behavior.	More SQL calls, failed attempts, fallback, or results that diverge from the selected component.
Data-limitation handling	The requested fact is not supported, but the model substitutes an undocumented proxy instead of qualifying or declining the claim.	A confident answer despite a semantic limitation or missing mapping.

Common cause	What goes wrong	Typical trace signature
Relationship and completeness	The calculation is partly correct, but identifiers are not resolved to labels or required subgroups, counts, qualifiers, or comparisons are omitted.	Correct rows or values paired with incomplete final evidence.
Units, precision, and evaluator boundary	The underlying result is close or substantively correct, but scaling, rounding, units, formatting, or the evaluator contract causes a strict miss.	Ratio-versus-percentage errors, out-of-tolerance rounding, or malformed judge output.

4.1 Stage-specific interpretation

- **Low discovery or resolution points upstream.** Improve component coverage, ranking, or detail loading.
- **High utilization with a wrong answer points downstream.** Inspect definition choice, SQL semantics, and answer assembly before adding more retrieval.
- Fallback paired with failed SQL signals a compatibility or query-validation problem, not automatically a missing definition.
- Correct execution with a strict miss requires a separate review of completeness, units, precision, and evaluator behavior.

4.2 Findings from the leading configurations

Opus 5 passed 46 of 48 semantic attempts. Its two misses occurred on the same task: both executions matched five of seven required assertions but omitted two required values. The failure mechanism was final-answer completeness, not semantic discovery or component coverage.

Sonnet 5 passed 27 of 48 semantic attempts. Its most common strict misses also involved omitted evidence, while its high raw-schema fallback rate indicates that access to semantic components did not consistently translate into reliance on them. This contrasts with Opus 5 and illustrates why retrieval, utilization, fallback, and answer completeness must be read together.

Execution errors were uncommon relative to the full matrix and were concentrated in timeout behavior. They remain part of the reliability outcome, while the broader failure pattern is dominated by analytical interpretation, query construction, and answer composition rather than infrastructure failure.

Overall, the semantic layer removes much of the discovery burden but does not by itself enforce analytical policy, query correctness, or answer completeness. A failure should therefore be assigned to a process stage before retrieval is changed or another semantic component is authored.

5. Discussion

The paired results provide evidence that semantic-layer access can improve analytical performance while reducing database exploration. Strict pass rate increased for 9 of 9 evaluated models, and SQL calls declined for every model. The cost result was less uniform: normalized model spend declined for 7 of 9 configurations because token-class composition and provider pricing differed materially.

The cross-model variation is itself a central result. Identical tasks and semantic context produced different patterns of retrieval, fallback, query retries, token use, and final-answer quality. The appropriate deployment unit is therefore the model–semantic-layer configuration, not the model or semantic layer considered independently.

Failure review further narrows the engineering interpretation. Relevant components were often available and sometimes heavily used in failed cases. Additional semantic coverage is therefore not a universal remedy;

downstream controls for canonical definition selection, SQL semantics, documented data limitations, relationship resolution, units, precision, and answer completeness are also required.

5.1 Descriptive correlates of semantic-mode pass rate

To identify factors associated with quality after the semantic layer is in place, this analysis uses one semantic-mode aggregate per model. Raw-schema observations and raw-to-semantic changes are excluded; the outcome is semantic-mode strict pass rate and every candidate metric is measured in the semantic condition. The aggregation also avoids treating the 48 task attempts within a model as independent observations. Spearman rho describes rank-order association across model means and is less sensitive than a linear correlation to extreme values and uneven metric scales. The coefficients are exploratory effect descriptions only: the model count is small, observations differ by provider and configuration, and no hypothesis tests or causal claims are made.

Table 7. Semantic-only cross-model correlation with strict pass rate, ordered by Spearman ρ from lowest to highest.

Candidate metric	Spearman ρ	Observed rank direction
SQL calls per case	-0.81	Higher metric, lower pass rate
Failed SQL calls per case	-0.79	Higher metric, lower pass rate
Total tokens per case	-0.75	Higher metric, lower pass rate
Schema fallback rate	-0.56	Higher metric, lower pass rate
Discovery precision	-0.37	Higher metric, lower pass rate
Payload utilization	-0.24	Higher metric, lower pass rate
Relationship usage	-0.19	Higher metric, lower pass rate
Used semantic objects per case	+0.14	Higher metric, higher pass rate
Cache-read tokens per case	+0.21	Higher metric, higher pass rate
Resolved-object utilization	+0.27	Higher metric, higher pass rate
Estimated model cost per case	+0.30	Higher metric, higher pass rate

The strongest observed rank associations were fewer SQL calls per case ($\rho = -0.81$), fewer failed SQL calls per case ($\rho = -0.79$), and fewer total tokens per case ($\rho = -0.75$). Fallback was also inversely associated with pass rate ($\rho = -0.56$). Discovery precision, cache-read volume, and cost per case were materially weaker correlates in this cross-model comparison.

Across the two editions, the more consistent signal is operational trajectory after retrieval: models with fewer query attempts, fewer failed queries, lower token use, and less fallback tended to rank higher on pass rate. Because the comparison is confined to semantic-mode aggregates, the SQL association reflects differences among models operating with the semantic layer, not a raw-versus-semantic mode contrast. Retrieval-volume measures alone were less diagnostic. The pattern is consistent with the trace review, in which many residual failures occurred after relevant semantic context had already been discovered.

6. Limitations

- **Run count.** Each model-condition pair was executed three times. This supports descriptive estimates of repeatability but remains insufficient for population-level significance claims.
- **Task and data scope.** The cohort contains 16 tasks on one analytical connection. External validity across domains, schemas, semantic-layer designs, and task distributions has not been established.
- **Automated evaluation.** Strict pass rate depends on typed targets, numeric tolerances, required evidence, and evaluator behavior. Qualitative review identifies boundary cases but does not retroactively change recorded scores.

- **Execution errors and imputation.** Seven timeouts and one malformed output remain non-passes. Missing additive usage is estimated from identical successful repetitions; one Sonnet 5 raw-schema estimate has only one donor and is sensitivity-only.

7. Conclusion

Within this controlled workload, semantic-layer access improved strict answer quality and reduced direct SQL work for every evaluated model. It also reduced normalized model cost for most configurations, although token and cost effects were not universal. The semantic layer did not eliminate failures after retrieval. These results support semantic layers as an effective system intervention, provided that each model-layer pair is evaluated jointly across quality, operational activity, normalized cost, retrieval behavior, and failure stage.

The three-trial design shows that the direction of the quality effect is consistent across models while its magnitude and operating mechanism vary substantially. Production qualification should therefore preserve both the repeated outcome measures and the process diagnostics that explain them.

References

ARC Prize Foundation. (2026). ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence. https://arcprize.org/media/ARC_AGI_3_Technical_Report.pdf

Bigeard, A., Nashold, L., Krishnan, R., & Wu, S. (2025). Finance Agent Benchmark: Benchmarking LLMs on Real-world Financial Research Tasks. arXiv:2508.00828.

Jimenez, C. E., Yang, J., Wettig, A., et al. (2023). SWE-bench: Can Language Models Resolve Real-World GitHub Issues? arXiv:2310.06770.

Kwa, T., West, B., Becker, J., et al. (2025). Measuring AI Ability to Complete Long Tasks. arXiv:2503.14499.

Starace, G., Jaffe, O., Sherburn, D., et al. (2025). PaperBench: Evaluating AI's Ability to Replicate AI Research. arXiv:2504.01848.

Vals AI. (2026). Finance Agent v2 benchmark report. <https://www.vals.ai/benchmarks/fabv2>

Pricing references

OpenAI Sol pricing. <https://developers.openai.com/api/docs/models/gpt-5.6-sol>

OpenAI Terra pricing. <https://developers.openai.com/api/docs/models/gpt-5.6-terra>

OpenAI Luna pricing. <https://developers.openai.com/api/docs/models/gpt-5.6-luna>

Anthropic list prices. <https://platform.claude.com/docs/en/about-claude/pricing>

Z.AI pricing. <https://docs.z.ai/guides/overview/pricing>

Kimi K2.5 pricing. <https://platform.kimi.ai/docs/pricing/chat-k25>